

C++ Standard Library – Full Summary (Syntax + Usage)

NOTE: This PDF contains a structured overview of the full C++ Standard Library: key headers, containers, algorithms, and usage patterns. Each entry provides syntax and a short description of how it works.

1. IOSTREAM

Syntax:

```
#include  
cin >> x;  
cout << x;
```

Use:

Input/output stream operations.

2. STRING

Syntax:

```
#include  
string s = "Hello";  
s.size();  
s.substr(0,3);
```

Use:

Text handling.

3. VECTOR

Syntax:

```
#include  
vector v;  
v.push_back(5);
```

Use:

Dynamic array.

4. ALGORITHM

Syntax:

```
#include  
sort(v.begin(), v.end());  
reverse(v.begin(), v.end());
```

Use:

Ready-made logic functions.

5. MAP

Syntax:

```
#include  
map mp;  
mp[1] = "A";
```

Use:

Key–value store (sorted).

6. UNORDERED_MAP

Syntax:

```
#include  
unordered_map mp;
```

Use:

Fast hash map.

7. SET

Syntax:

```
#include  
set st;  
st.insert(10);
```

Use:

Sorted, unique values.

8. STACK

Syntax:

```
#include  
stack st;
```

Use:

LIFO structure.

9. QUEUE

Syntax:

```
#include  
queue q;
```

Use:

FIFO structure.

10. PRIORITY QUEUE

Syntax:

```
#include
```

priority_queue pq;

Use:
Highest priority element first.

11. DEQUE

Syntax:
#include
deque dq;

Use:
Adds/removes from front/back.

12. LIST

Syntax:
#include
list L;

Use:
Doubly linked list.

13. TUPLE

Syntax:
#include
tuple t;

Use:
Store multiple types.

14. PAIR

Syntax:
#include
pair p;

Use:
Two-item container.

15. BITSET

Syntax:
#include
bitset<8> b;

Use:
Efficient fixed-size bits.

16. THREAD

Syntax:

```
#include  
thread t(func);
```

Use:

Multithreading.

17. MUTEX

Syntax:

```
#include  
mutex m;
```

Use:

Thread locking.

18. FILESYSTEM

Syntax:

```
#include  
filesystem::exists("file.txt");
```

Use:

File operations.

19. MEMORY (Smart Pointers)

Syntax:

```
#include  
unique_ptr p = make_unique(5);
```

Use:

Safe dynamic memory.

20. OPTIONAL / VARIANT / ANY

Syntax:

```
optional x;  
variant v;
```

Use:

Modern type containers.

21. MATH (CMATH)

Syntax:

```
#include  
sqrt(9);  
pow(2,5);
```

Use:
Math operations.

22. CHRONO

Syntax:
`#include`
`auto t = chrono::high_resolution_clock::now();`

Use:
Time measurement.

23. FSTREAM

Syntax:
`#include`
`ofstream f("out.txt");`

Use:
File read/write.

24. EXCEPTION HANDLING

Syntax:
`try { ... }`
`catch(exception& e) { ... }`

Use:
Error handling.

This PDF gives a clean map of essential components.
Ask anytime if you want a deeper expanded version or full examples for each item.